A family of defining quartics in 263 variables

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Abstract

Under the stated arithmetic premises of the elliptic source, we specify an integer quartic family in 263 total variables whose sufficiently large members define the integers in the rationals. Four disjoint affine images of the contact set give a height exponent 19/4. This permits quadratic Taylor approximations to the formal logarithm, with comparison orders twelve and three. Residue classes modulo eight permit sixteen local tests to be represented by five factors. A contact set symmetric about one makes its degree-twenty product a degree-ten polynomial in the squared displacement. Five quadratic coordinates evaluate that product and its normalization. The normalization is independent of the scale and uses the positive aggregate trace directly. The complete quadratic and expanded coefficient tables are independently verified. The family has 3312 monomials for every positive parameter, and exact coefficient height $80989^{4N+2}+1$ for $N \geq 23$. The sufficient defining parameter remains ineffective because the aggregate-trace height theorem has an ineffective additive constant.

1 The family and its arithmetic scope

All variable counts include the distinguished input X. For a fixed integer $N \geq 1$ let $A_N = 80989^{2N+1}$. The complete tables `quadratic-family263.json` and `polynomial-family263.json` specify 237 primitive integer quadratic polynomials and their sum of squares

$$P_N(X, Y_1, \ldots, Y_{262}) = \sum_{j=1}^{237} Q_{N,j}^2.$$

The parameter A_N is a coefficient, not a variable.

Theorem 1.1. *Assume the arithmetic premises stated below. There is a fixed integer $N_0 \geq 27464$ such that every $N \geq N_0$ satisfies*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Q}^{262}\ P_N(x, \mathbf{y}) = 0 \qquad (x \in \mathbb{Q}).$$

For every $N \geq 1$ the polynomial P_N has degree exactly four, content one, and 3312 nonzero coordinate monomials. For $N \geq 23$,

$$H(P_N) = 80989^{4N+2} + 1.$$

The construction specifies every member exactly. The proof does not determine a sufficient numerical value of N_0.

Set

$$K = \mathbb{Q}(\rho), \quad \rho^2 = 2, \quad D = (3-\rho)^2 = 11 - 6\rho, \qquad E : y^2 = x^3 - Dx,$$
$$P = 2(1+\rho, 2\rho) = \left(\frac{57 - 28\rho}{8}, \frac{-660 + 417\rho}{32} \right).$$

We retain the following arithmetic premises, with the precise domains and hypotheses recorded in [1, 2, 3]:

1. $2E(K) = \mathbb{Z}P$ and $h(x([n]P)) \geq n^2 - 6$ for every nonzero integer n, together with the specified point charts and group domains.

2. The five-contact height estimate with exponent nineteen, exceptional cutoff 76176, and the fixed constant
$$H = 2^{110}210^{1800}\left(\prod_{p\leq 76176} p\right)^{19} + 128.$$

3. The pole obstruction outside $2, 3, 17$, and its simultaneous realization for positive inputs less than 17 with square reduced denominator and the stated local squareclasses and valuation bounds at those three primes. Its normalized quadratic equations are retained.

4. The simultaneous prime-pattern, conjugate reduction-order, finite congruence, and real approximation assertions used by the source. The weighted selection and fixed denominator dilation are justified from the cited analytic estimates in [7].

The aggregate trace-height theorem used below is the proved consequence of the family height theorem described in [6]; its additive constant is ineffective. The contact amplification in [5] is an elementary consequence of the retained five-contact estimate, not an additional arithmetic assumption. Exact coefficient verification does not prove the retained arithmetic premises.

2 Four contact blocks and the two scales

Put

$$\Theta = \{0, -1/2, -1, 1/2399, -32/407\}, \qquad \Theta_4 = \Theta \cup (2 - \Theta) \cup (\Theta + 5) \cup (-3 - \Theta),$$
$$F_4(t) = \prod_{\theta\in\Theta_4} (t - \theta).$$

The twenty contacts are distinct, omit one, and are symmetric about one. Let Σ consist of the primes at most 76176 and the three primes $247957, 368323, 2029147$. Outside this set, all contact differences and all numbers $1-\theta$ are units. Define $M_4(t)$ as the product of primes $p \notin \Sigma$ for which $v_p(t-\theta)$ is positive and odd for some $\theta \in \Theta_4$. Exact factorizations in the independent certificate verify the exceptional set. They also show $M_4(1) = 1$.

Put $E_4 = 247957 \cdot 368323 \cdot 2029147$. Then

$$h(t) \leq \log 14 + HE_4^{19} M_4(t)^{19/4} \leq H_4 M_4(t)^{19/4}, \qquad H_4 = \log 14 + HE_4^{19}, \tag{1}$$

for $t \notin \Theta_4$. Here is a direct proof from the retained five-contact inequality. Apply it to the four maps
$$2 + \frac{1}{t}, \qquad 2 + \frac{1}{2-t}, \qquad 2 + \frac{1}{t-5}, \qquad 2 + \frac{1}{-3-t}.$$
Their integer matrices have determinant ± 1. The maximum absolute row sums of inverse projective matrices are respectively $3, 7, 14, 8$, so each inverse changes logarithmic height by at most $\log 14$. Let M_j be the contact radical for the corresponding five-element block, omitting Σ. Its original radical is at most $E_4 M_j$. Thus $h(t) - \log 14 \leq HE_4^{19} M_j^{19}$ for each of the four blocks. If the left side is positive, multiplication gives the first inequality in (1), since the block radicals are pairwise coprime and their product is $M_4(t)$. If it is nonpositive, the inequality is immediate. This accounts for every fixed exceptional prime.

The retained root points are

$$Q_1 = [n]P, \quad Q_2 = [a]P, \quad Q_3 = [t]P, \quad Q_4 = [b]P, \qquad B = \max(|n|, |a|, |t|, |b|),$$

with nonzero integer labels. Set

$$T = \frac{1}{4}\operatorname{Tr}_{K/\mathbb{Q}}\left(\sum_{i=1}^{4} x(Q_i)\right).$$

It is positive, and the aggregate trace theorem gives fixed constants $M_1 > 0$, $M_2 \geq 1$ such that

$$B^2 \leq M_1 h(T) + M_2. \tag{2}$$

Use the two positive rational scales $a_* = 1, T$. No square condition on a scale is needed. Replacing the half-trace in the aggregate height theorem by its half changes its logarithmic height by at most $\log 2$; this is absorbed into M_2. For each scale put $t_* = a_*\lambda$ and $\delta = t_* - 1$. On the nonzero domain the detector is

$$R = \frac{F_4(t_*)}{\delta^{22}}, \qquad q = \frac{\delta^{18}}{F_4(t_*)} = \frac{1}{R\delta^4}. \tag{3}$$

The contact circuit stores $r = \delta^2$ and its powers r^2, r^4, r^5. The actual quadratic normalization equations are

$$F_4(t_*)q = r^4 r^5, \qquad k = qr^2. \tag{4}$$

Thus $k = R^{-1}$ when $\delta \neq 0$. No contact value can occur: $F_4(t_*) = 0$ would force $r = 0$ and $t_* = 1$, contradicting $F_4(1) \neq 0$. If $q = 0$, the same equation forces $\delta = 0$. At $\delta = 0$ the contact radical is one; no reciprocal is used on this stratum.

If both radicals divide a nonzero integer J to order k, then $h(T) \leq h(\lambda) + h(T\lambda)$ and (1)–(2) imply

$$B^2 \leq C|J|^{19/(4k)} + M_2, \qquad C = 2M_1 H_4. \tag{5}$$

The two radicals need not be coprime for this implication.

3 Local tests and the approximation denominators

Let σ be conjugation, $f = 1/x$, and $z = -x/y$, extended by zero at the identity where these functions are regular. The retained point relations are

$$U_0 = H, \quad V_0 = H - P, \quad U_i = [u_i]P, \quad V_i = U_i - Q_i.$$

The charts permit $U_i = O$ or $V_i = O$ for $i > 0$. Put

$$G_2(w) = 1 - \frac{2}{5}w - \frac{2}{15}w^2, \quad \ell_0 = z(V_0)G_2(Df(V_0)^2), \quad L_i = \frac{z(U_i)}{z(H)},$$

$$L_1' = \sigma\left(\frac{z(V_1)}{\ell_0}\right), \qquad L_i' = \sigma\left(\frac{z(V_i)G_2(Df(V_i)^2)}{\ell_0}\right) \quad (i > 1).$$

These quotients are interpreted only after the denominator exclusions below. Write

$$D_H = f(H), \quad A_i = f(U_i), \quad B_i = \sigma f(V_i), \quad C_i = L_i - L_i', \quad I = L_1 - \beta, \quad \beta = 51X^2 + 2504,$$

$$G = L_3^2 - L_2(L_2 - 2)L_3L_4 + (L_2 - 2)^2 L_4^2 - 1.$$

The required orders are six for D_H, twelve for C_2, C_3, C_4, G, three for C_1, six for I, and one for the nine A_i, B_i. The final input comparison uses order three, the minimum of the two orders controlling $L_1 - n$ and $L_1 - \beta$.

Lemma 3.1. *The polynomial G_2 has no zero in K.*

Proof. Its roots are $(-3 \pm \sqrt{39})/2$, and 39 is neither a rational square nor twice a rational square. Explicitly, if $(u + v\sqrt{2})^2 = 39$ with $u, v \in \mathbb{Q}$, then $uv = 0$; the two cases would require $u^2 = 39$ or $v^2 = 39/2$, both impossible. The independent finite certificate also verifies that the integer numerator $15 - 6w - 2w^2$ has no root modulo 17, a prime that splits in K. Its leading coefficient is a unit there, so a hypothetical local root would be integral and give a forbidden residue root. □

The finite H chart is $xf = 1$, $S^2 = x - Df$, $H = (x, xS)$. The source group calculation writes $H = [m]R_0 + T_0$ with $P = 2R_0$, $m \in \mathbb{Z}$, and $T_0 \in E[2](K)$. Put $h = m/2$ for its rational free index. The exceptional classes $E[2](K), \pm P + E[2](K)$ are excluded before any logarithmic ratio is interpreted. Indeed their finite allowed reciprocal abscissas are units outside the fixed small primes, whereas extraction below would require positive valuation at any contact. All radicals would then be one, forcing $B^2 \le C + M_2$ by (5). The retained scalar lower bound in Section 5 contradicts this for every sufficiently large parameter. The explicit torsion translations and norm values supporting the unit assertion are those recorded in [4, Section 3]. This exclusion uses no comparison quotient.

Consequently $m \notin \{-2, 0, 2\}$, and H, V_0 are nontorsion. Their required nonzero abscissas and ordinates, and all subtraction denominators, follow from the retained chart identities. In particular $z(H), z(V_0) \neq 0$. Lemma 3.1 gives $\ell_0 \neq 0$ globally. The analytic formal logarithms have a separate nonvanishing proof from the formal logarithm isometry when the points are formal.

4 Five factors and extraction at one place

At a contact prime $p \notin \Sigma$, δ is a unit and $\nu = v_p(R) = v_p(F_4(t_*))$ is positive and odd. This follows from the unique contact factor with positive odd valuation and does not require any condition on the valuation of the scale. All valuations of K are unramified and normalized by $v(p) = 1$. For a point on E, every finite value of $v(f)$ is even: if $v(x) < 0$ then $2v(y) = 3v(x)$, while if $v(x) > 0$ then $2v(y) = v(x)$; the remaining case is valuation zero. This also holds for conjugated reciprocal abscissas. Before excluding the exceptional H classes, the stored value for $f(H - P)$ still has even valuation: its literal equation is $f(H)F_0 = g^2$, and $f(H)$ has even valuation. Thus the parity argument and the preceding exclusion do not assume the validity of a division in the fixed subtraction chart.

Use

$$
\begin{aligned}
T_1 &= 1 + q^{23}C_2^2 + q^{10}D_H^2 + q^4B_1^4,\\
T_2 &= 1 + q^{23}G^2 + q^2A_3^2 + q^4B_0^4,\\
T_3 &= 1 + q^{23}I^4 + qA_2^2 + q^2A_4^2 + q^4B_3^4,\\
T_4 &= 1 + q^{23}C_3^2 + q^{10}C_1^4 + q^4B_4^4,\\
T_5 &= 1 + q^{23}C_4^2 + q^2A_1^2 + q^4B_2^4, \qquad W = T_1T_2T_3T_4T_5.
\end{aligned}
$$

Lemma 4.1. *At a contact prime, every factor is nonzero and has valuation at most zero. A factor of valuation zero is congruent to one in the residue field.*

Proof. First suppose every displayed test in a factor is nonzero. In each factor other than T_3, the three nonconstant valuations belong, respectively, to the odd integers, to $2 + 4\mathbb{Z}$, and to $4 + 8\mathbb{Z}$. For example T_1 has valuations

$$2v(C_2) - 23\nu, \qquad 2v(D_H) - 10\nu, \qquad 4v(B_1) - 4\nu.$$

The second has exact factor two because $v(D_H)$ is even and ν is odd. The third has exact factor four because $v(B_1)$ is even. The term $q^{10}C_1^4$ in T_4 has valuation $2(2v(C_1) - 5\nu)$, again twice an odd integer.

In T_3 the four nonconstant valuations are

$$4v(I) - 23\nu, \quad 2v(A_2) - \nu, \quad 2v(A_4) - 2\nu, \quad 4v(B_3) - 4\nu.$$

The first three have residues $\nu, -\nu, 2$ modulo four. These are pairwise distinct and nonzero because ν is odd and the A_i have even valuations. The fourth has residue four modulo eight; it therefore differs from each of the first three and from zero. Thus, in every factor, all finite term valuations, including zero for the constant term, are distinct. Zero tests simply remove terms from this argument. The minimum is attained exactly once. This proves nonvanishing and the asserted bound. If the minimum is zero, every nonconstant term has positive valuation, so the residue is one. □

Proposition 4.2. *At each contact prime, the retained pole system and the cleared trace equations imply $\nu > 0$ and give all sixteen required test orders at one place above that prime.*

Proof. Write $W = w_0 + w_1\rho$. The literal cleared trace equations are

$$w_0^2 - 2w_0\tau - 2w_1^2 = 0, \qquad k\mathfrak{b} = \tau^2, \qquad k = R^{-1}.$$

By Lemma 4.1, $W \neq 0$. The first equation therefore gives $w_0 \neq 0$ and $\tau = 1/\operatorname{Tr}(W^{-1}) \neq 0$; indeed $w_0 = 0$ would give $w_1 = 0$. Hence $\mathfrak{b} = R/\operatorname{Tr}(W^{-1})^2$. If $\ell = \max_{v|p} v(W) \leq 0$, the trace inequality gives $v_p(\mathfrak{b}) \leq \nu + 2\ell$. This valuation is odd; the retained pole obstruction forces it positive. Consequently $\nu > 0$ and $\ell > -\nu/2$. At a place attaining ℓ, the factor valuations are nonpositive and add to ℓ. Each factor valuation is therefore at least ℓ, and each of its term valuations is strictly greater than $-\nu/2$.

The resulting lower bounds, before integer rounding, are

test	strict lower bound	integer order
C_2, C_3, C_4, G	$45\nu/4$	12
I	$45\nu/8$	6
C_1	$19\nu/8$	3
D_H	$19\nu/4$	6
A_2	$\nu/4$	1
A_1, A_3, A_4	$3\nu/4$	1
B_0, B_1, B_2, B_3, B_4	$7\nu/8$	1

Here ν is a positive odd integer. The order six for D_H uses its even valuation. In fact each of the nine reciprocal-abscissa tests has order at least two; order one suffices for its subsequent formal-group use. Zero tests satisfy all the stated inequalities. □

4.1 Integrality precedes the depth estimate

The extracted positive f valuations put H, U_i on the primary formal branch and V_0, V_i on the conjugate formal branch. At a nonidentity formal point of depth $r = v(z) \geq 1$, write

$$\log_E(Q) = z(Q)\mathcal{A}(Q), \qquad \mathcal{A}(Q) = G(Df(Q)^2), \qquad \mathcal{A}(O) = 1.$$

The formal series satisfies

$$G(w) = 1 - \frac{2}{5}w - \frac{2}{15}w^2 + O(w^3), \qquad g_j = \frac{4j-5}{4j+1}g_{j-1} \quad (j \geq 2).$$

The coefficient bound and full-tail proof in [4, Lemma 3.1] give $\mathcal{A} \equiv 1 \pmod{p^{4r}}$, $G_2/\mathcal{A} \equiv 1 \pmod{p^{12r}}$, and $1/\mathcal{A} \equiv 1 \pmod{p^{4r}}$. These analytic units equal one at O, so zero numerator points require no quotient $\log_E(O)/z(O)$.

The integral group identities

$$[m]U_i = [2u_i]H, \qquad [m-2]V_i = [2(u_i - n_i)]V_0$$

kill the torsion component of H and yield full logarithm ratios $r_i = u_i/h$ and $r_i' = (u_i - n_i)/(h-1)$. At the odd prime, $h, h-1$ are integral and at least one is a unit. Hence at least one full ratio is integral. Multiplication by the analytic units preserves the corresponding approximate valuation. The positive-order cross test transfers integrality to the other approximate ratio and thus to both full ratios.

Only now use $L_i = z(U_i)/z(H)$ integral. The H test gives $v(z(H)) \geq 3$, so every nonzero U_i has depth at least three. The primary errors have order at least twelve. The alternate errors have order at least four for $i = 1$ and twelve for $i > 1$. The integral full ratios convert these relative bounds into absolute congruences. Multiplication by $h, h-1$ and subtraction then give

$$L_1 \equiv n \pmod{p^3}, \qquad L_i \equiv n_i \pmod{p^{12}} \quad (i > 1).$$

This argument includes zero U_i or V_i, half-integral h, and inert primes. At the inert primes conjugation preserves the valuation.

5 Global soundness for sufficiently large parameters

The retained scalar circuit gives, for $c = |a - 1|$,

$$c \geq 80989^{L_N}, \qquad L_N = \left\lfloor \frac{187(2N-1)}{8} \right\rfloor + 1, \qquad h(\beta) \leq \mathcal{H} c^{19/24}. \tag{6}$$

These statements, including their integral group identities and exceptional-point exclusions, are the scalar conclusions of [2]; no comparison logarithm is used to establish them. Their circuit is retained, including the actual positive contact product r and the normalizer

$$r = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1), \qquad R_s = \frac{A_N r}{1 + 80989^2 r}.$$

Let $\Delta = t^2 - a(a-2)tb + (a-2)^2 b^2 - 1$. The local comparison shows that each twelfth power of a comparison radical divides Δ. If $\Delta \neq 0$, then $|\Delta| \leq 14B^4$, and (5) gives

$$B^2 \leq C 14^{19/48} B^{19/12} + M_2.$$

Since $19/12 < 2$, this bounds B by a fixed constant. Equation (6) gives $B \geq c - 1 \to \infty$ uniformly as $N \to \infty$, a contradiction. Thus $\Delta = 0$ eventually. The retained integral Lucas descent then gives

$$\log B > |a| - 3 \geq c - 4.$$

If $n - \beta \neq 0$, let J be its nonzero reduced numerator. The local input comparisons make both cubed radicals divide J, while $|J| \leq 2B \exp(h(\beta))$. For $B^2 \geq 2M_2$, (5) gives

$$\log B \leq \frac{12}{5}\log(2C) + \frac{19}{5}\log 2 + \frac{19}{5}h(\beta) \leq C' + \frac{19}{5}\mathcal{H} c^{19/24}.$$

The right side is $o(c)$, contradicting the preceding lower bound. Hence $n = \beta \in \mathbb{Z}$. If $X = u/v$ is reduced with $v > 0$, then $v^2 \mid 51$, and squarefreeness of 51 gives $v = 1$. All bounds depend only on fixed constants. Since L_N is increasing, a single N_0 works for every larger N. Its dependence on the aggregate additive constant is ineffective.

6 Simultaneous completeness

Fix an integer input and the coefficient parameter. Retain the scalar and Lucas construction of the source, and choose the integer labels, root points, two scales, and compact real and finite-place neighborhoods before selecting primes. The neighborhoods avoid every retained rational denominator; Lemma 3.1 ensures that the new denominator introduces none. Every local test then has a fixed lower valuation bound at each prime in the finite preparation set.

Choose a fixed positive integer D_0 clearing all contact coefficients $D_0^2\theta/a_*$ and sufficiently divisible at the prepared primes, including Σ, and the prime divisors of the scales. In the prime pattern write

$$\lambda = \frac{u}{D_0^2 w^2}.$$

The zero contact makes u one of the selected primes, outside that finite set. Thus $v_p(\lambda) \le -2v_p(D_0)$ there. Make this upper bound low enough that $t_* = a_*\lambda$ has valuation below zero and all nonzero contacts. Then $v_p(\delta) = v_p(t_*)$, and the exact valuation formulas are

$$v_p(F_4(t_*)) = 20v_p(t_*), \qquad v_p(q) = v_p(R) = -2v_p(t_*).$$

The last two valuations become uniformly arbitrarily large. Thus all weighted test terms are small at every prepared place, and $\mathrm{Tr}(W^{-1})$ is arbitrarily close to two.

The local squareclasses follow directly from

$$R = \delta^{-2} \prod_{\theta\in\Theta_4} \left(1 + \frac{1-\theta}{\delta}\right). \tag{7}$$

The factors in the product are arbitrarily close to one. Requiring congruence to one modulo eight at 2, and modulo p at an odd prepared prime p, makes their product a local square. The prefactor δ^{-2} is a square, as is the reciprocal trace factor in the pole input. Taking $-v_p(t_*)$ sufficiently large also gives the required pole valuations at $2, 3, 17$, including the subtraction of $2v_p(2)$ in the trace denominator.

This preparation is valid for any fixed nonzero rational scales. The prime-pattern input in [7, 1] applies to arbitrary fixed distinct integer coefficients. After clearing $D_0^2\theta/a_*$ and removing repeated coefficients, it applies here without a square restriction on a_*. All denominator conditions are fixed before prime selection.

Select the enlarged finite set of contact primes with the inherited opposite-order separation. The common-index construction takes $H = [h]P$, $U_i = [hn_i]P$, and $V_i = [(h-1)n_i]P$. Its congruences retain primary depth twenty for H and every nonzero U_i. Thus the full primary logarithm ratio is exactly n_i, and its parameter-ratio approximation has error of order eighty. In particular, since $n_1 = \beta$ in the completeness construction, $I = L_1 - \beta$ has order eighty; the stronger requirement six on this test imposes no new arithmetic condition. At the opposite places the points are formal. The new Taylor approximations give comparison precision twelve, and the constant approximation at V_1 gives precision four. The reciprocal-abscissa tests have valuation at least two. At the chosen primary place, where $v_p(R) = 1$, it therefore suffices to use orders $6, 12, 4, 12$ respectively for D_H, the high comparisons, C_1, and I: every nonconstant term in all five factors has positive valuation. All these orders follow from the preceding construction. There $W \equiv 1$. At the opposite place, the valuation-class argument gives either positive valuation for W^{-1} or residue one. Its trace is consequently a unit with residue one or two, including the inert case. Hence the pole input has valuation one at every selected contact prime.

The exact prime-pattern identity is

$$R = \frac{a_*^{20} \prod_{\theta\in\Theta_4} (u - (\theta/a_*)D_0^2 w^2)}{(D_0^2 w^2)^{20}\delta^{22}}.$$

Every factor outside the selected prime numerators is a rational square, including a_*^{20} even when a_* is not itself a square. Thus at every prime outside those numerators the valuation is even. Division by the squared trace preserves this parity. This argument includes primes dividing δ; no unit assertion about δ is made away from contacts. Equation (7) and the preparatory divisibility give the required local squareclasses and valuation bounds at $2, 3, 17$.

On the fixed compact real neighborhoods,

$$R \sim a_*^{-2}\lambda^{-2}, \qquad q \sim a_*^{-2}\lambda^{-2}, \qquad W \longrightarrow 1, \qquad \mathrm{Tr}(W^{-1}) \longrightarrow 2.$$

All test powers in the factors are even, so the pole input tends to zero through positive values. It eventually lies in $(0, 17)$. The simultaneous selection supplies the required real size after all fixed data have been chosen. The pole-realization input supplies its witnesses, and the exact power identities fill the quadratic circuit. This proves completeness with no additional condition on N.

7 Exact quadratic specification

The weights are specified by six scalar quadratic equations:

$$q_2 = q^2, \quad q_4 = q_2^2, \quad cq_4 = q, \quad q_5 = qq_4, \quad q_{10} = q_5^2, \quad cq_{23} = q_{10}^2.$$

At every contact $q \neq 0$, so $c = q^{-3}$ and each q_j is exactly q^j. The equations make no division by a variable. If $q = 0$, the clearing equations force $\delta = 0$ and the contact radical is one, as already proved. No value of the unconstrained reciprocal weight on that stratum affects the soundness argument.

Lemma 7.1. *The values $F_4(t_*)$, $(t_* - 1)^{18}$ and $(t_* - 1)^4$ are quadratic expressions in five additional coordinates, each determined by one successive quadratic equation over $\mathbb{Q}$.*

Proof. Put $r = (t_* - 1)^2$. Symmetry of the contacts gives

$$F_4(t_*) = P_{10}(r), \qquad P_{10}(r) = \prod_{\theta \in \Theta} (r - (\theta - 1)^2)(r - (\theta + 4)^2).$$

The ten squared roots are distinct. Four successive equations compute

$$r = (t_* - 1)^2, \quad r_2 = r^2, \quad r_4 = r_2^2, \quad r_5 = rr_4.$$

Every integer from zero through ten is a sum of two elements of $\{0, 1, 2, 4, 5\}$. Therefore $P_{10}(r)$ is a quadratic expression in $1, r, r_2, r_4, r_5$, and a fifth equation computes it. The other required values are already quadratic expressions:

$$(t_* - 1)^{18} = r_4 r_5, \qquad (t_* - 1)^4 = r_2.$$

All coefficients are fixed rational numbers. These equations determine the claimed values at every input, including the contact values and one. □

The normalization is consequently $F_4(t_*)q = r_4 r_5$, followed by $k = qr_2$. The trace-clearing coordinate is $k = R^{-1}$ on the nonzero domain. The second scale is an affine rational expression in the retained root-abscissa coordinates, so its product with λ is quadratic without a separate scale coordinate.

The high-point approximants G_2 are affine expressions in the retained f^2 and w^2 coordinates. They introduce no additional field value. At V_1 the numerator is simply $z(V_1)$. The shared field powers requiring additional coordinates are

$$C_1^4, \quad I^4, \quad B_1^4,$$

each with two rational coordinates. The fourth powers of B_0, B_2, B_3, B_4 are affine expressions in the Taylor circuits. Each detector has five field factors and three private intermediate products, using sixteen coordinates; the final product uses the retained W coordinates. The exact count is

point, scalar, contact, logarithm, and weight coordinates	225
three shared fourth powers	6
five factors and intermediate products for two detectors	32
total coordinates, including the input	263
point, scalar, contact, logarithm, and weight quadratic rows	195
shared fourth-power rows	6
factor and product rows	36
total quadratic rows	237

The complete tables fix every coordinate, equation, coefficient, and normalization; the input is coordinate one. The producer clears denominators and primitive-normalizes each row before squaring. An independent checker imports no producer. It recovers P_{10} by Newton finite differences at eleven exact arguments, checks all ten roots, and independently verifies $P_{10}((t-1)^2) = F_4(t)$ as a degree-twenty polynomial identity. It reconstructs every modified quadratic equation, compares the other equations with their pinned coefficient source, and checks the full contact exceptional set and all four height matrices. Both normalizations and the linear trace scale are verified directly. The projection proof uses these literal equations and the preceding arithmetic argument. It expands the entire sum of squares independently by unordered pairs of terms. The supplementary local certificate exhausts all odd residues of ν modulo eight and all possible valuation residues of every test. These complete residue classes verify the noncancellation assertion for all integer valuations, together with the exact extraction bounds and sufficient completeness depths. The infinite arithmetic arguments are the proofs above.

The expanded quartic has 3313 terms before powers of A_N belonging to the same coordinate monomial are collected, and 3312 monomials after collection. Its unique coefficient involving A_N^2 is $A_N^2 + 1$, at the square of the actual scalar contact-product coordinate. Every other coefficient is a nonzero constant or a nonzero constant multiple of A_N. Consequently no support cancellation occurs for positive A_N. The largest constant coefficient C_0 has 1520 bits, the largest linear multiplier is 13118436242, and exact comparison gives

$$C_0 > 13118436242^2, \qquad 80989^{90} + 1 < C_0 < 80989^{94} + 1.$$

Thus $H(P_N) = \max(C_0, A_N^2 + 1)$ for every $N \geq 1$ and the asserted exact height holds for $N \geq 23$. This is a coefficient statement; it does not make the defining threshold effective.

References

[1] Project repository, *Arithmetic dependencies of the defining-polynomial constructions*, `arithmetic_dependencies_20261108_v8.tex`.

[2] Project repository, *A family of quartics in 297 variables*, `quartic297_20261109_v12.tex`, especially Sections 2–6.

[3] Project repository, *Common local weights and quartics in 297 and 342 variables*, `shared_local_weights_20261109_v12.tex`.

[4] Project repository, *An explicit quartic in 322 variables*, `quartic322_20261109_v13.tex`.

[5] Project repository, *Height Bounds from Disjoint Sets of Rational Contacts*, `contact_height_amplification_20261109_v13.tex`.

[6] Project repository, *The height of an aggregate elliptic trace*, `aggregate_trace_height_20261108_v9.tex`.

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